

YASH RAJ PANDEY

AI Systems Engineer | LLM Infrastructure, Agent Platforms, Evaluation
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I ship correctness fixes into the open-source stack that other AI systems are built on: **69 merged pull requests across 30 projects**, including llama.cpp, RAGFlow, mem0, LiteLLM and MLX. Day to day I build and run a production LLM agent and retrieval platform at the University of Florida, promoted from Software Engineer to AI Agents Architect in 13 months. Most of that work is the part that proves an answer is right before a scientist acts on it.

Experience

University of Florida

AI Agents Architect

Gainesville, FL

Apr 2026 – Present

- Architected a production AI agent platform: retrieval over 10,000+ technical documents, tool calling, and natural-language access to research data.
- Replaced about 20 colliding rule-based routers with one deterministic intent arbiter, removing a class of wrong-tool failures the model could not detect.
- Built the release gate that decides what ships: 26 task evaluations and a 24-case regression suite across 14 candidate models. It caught hallucination failures before deployment.
- Led the migration of generation onto university-operated GPUs for a large open-weight model. Retrieval and reranking stay local, where a cross-encoder measured better.

Lead Software Engineer

Oct 2025 – Apr 2026

- Took over as lead of Blue Omics, the UF blueberry breeding program's genomics platform, when the previous lead left. System of record for 30+ researchers across 5 labs; grew the team from one to three.
- Rebuilt the backend and frontend as a second platform version: consolidated the data into one PostgreSQL schema of 30 Django models over 47 tables, loaded by 11 ingestion commands, then tuned it to 250 indexes over 18 GB with core lookups at 23–29 ms.
- Put every scientific tool behind one authenticated API, 75 routes covering gene search, BLAST, orthogroups, genome coordinates, transcriptome and metabolite data, plus an embedded JBrowse2 genome browser. Cut R Markdown reporting from 2–3 hours to 15–20 minutes.
- Found BLASTn silently running Megablast, so cross-species searches returned empty and read as real negatives. Switched to the sensitive nucleotide task and audited all 126 production BLAST databases.

Software Engineer

Mar 2025 – Oct 2025

- Joined Blue Omics at 81 source files and built the researcher-facing frontend: phenotype, genomics and transcriptome views, and the tool pages over BLAST and sequence search. Cut frontend load time from 8s to 3s.

Selected Projects

Open Source | 69 merged pull requests across 30 projects

github.com/devYRPauli

- **llama.cpp** (126k stars), **RAGFlow** (89k), **mem0** (64k), **LiteLLM** (57k) and **MLX** (28k): correctness fixes in inference kernels, document parsers, vector retrieval and per-request billing. Each begins with a reproduced failure and ends with a regression test.

willitcall | A compatibility matrix for tool calling on local models

github.com/devYRPauli/willitcall

- Rust CLI that runs one conformance suite against any OpenAI-compatible endpoint, publishing 32 rows across 3 serving stacks. Nobody had published the matrix.

Looma | Local-first memory for coding agents

github.com/devYRPauli/looma

- Zero-dependency Python CLI turning Claude Code, Codex and Cursor transcripts into resumable context. F1 0.86, 134 tests, published on PyPI.

Independent Evaluations | TurboQuant KV-cache compression and Google TabFM

yashrajpandey.com/writing

- Restored TurboQuant needle retrieval from 0% to 100% at 16K on a 16 GB M1 Pro by root-causing five bugs. Landed a multi-GPU crash fix in google-research/tabfm.

Technical Skills

AI systems: LLM inference and evaluation, RAG, agent runtimes, tool calling, vLLM, llama.cpp, MLX, quantization, vector search, reranking, MCP

Languages: Python, TypeScript, JavaScript, SQL, C/C++, Rust, Bash

Frameworks and data: Django, FastAPI, React, Next.js, PostgreSQL, DuckDB, SQLite, Redis, Qdrant, Supabase

Infrastructure: Docker, Linux, GPU and self-hosted inference, GCP, Vercel, Git, GitHub Actions

Education

University of Florida – M.S. Computer & Information Science & Engineering (GPA 3.8)

Aug 2023 – Dec 2024

Jaypee University of Engineering and Technology – B.Tech. Computer Science

Jul 2019 – May 2023